

YANAN HE

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🌐 <https://echoo113.github.io>

RESEARCH INTERESTS

- **Methodology:** foundation models, trustworthy and interpretable machine learning.
- **Applications:** natural language processing, AI for science.

EDUCATION

Purdue University, West Lafayette, IN

Aug 2023 – May 2026

Bachelor of Science in Computer Science & Mathematics-Statistics

GPA: 3.96/4.0

Awards: Dean's List (2023–2025), Semester Honor (2023–2025)

Relevant Coursework:

AI & CS: Machine Learning, Data Mining, Intro to AI, Algorithms Analysis, Data Structures, System Programming, Computer Architecture

Math & Stats: Numerical Methods, Stochastic Processes, Linear Algebra, Probability, Statistical Inference, Real Analysis

RESEARCH EXPERIENCE

Stony Brook University, Stony Brook, NY

May 2025 – Present

Research Assistant | Machine Learning

Advisor: Tengfei Ma

Project I: MTS-JEPA: Multi-Resolution Joint-Embedding Predictive Architecture for Time-Series Anomaly Prediction

- Led development of a self-supervised framework for time-series anomaly prediction, introducing a **multiscale JEPA-style architecture** to model long-range temporal dependencies.
- Addressed the **JEPA representation collapse** problem via a **soft-quantized codebook latent space**, improving stability of non-contrastive training and representation consistency.
- Achieved **state-of-the-art performance** on multiple benchmarks, improving **AUC and F1 by up to 5%**; manuscript submitted to **ICML 2026**.

Project II: Walking in the Ball: Hyperbolic Latent Navigation for Structured Thought Generation

- Leading a research project on hyperbolic latent navigation for **large language model reasoning**, exploring how non-Euclidean geometry can better model hierarchical reasoning structures.
- Introduced a geometric reasoning framework that embeds latent thought vectors in a **Poincaré ball** and models reasoning trajectories through **geodesic flows in hyperbolic space**.
- Designing the training framework and experimental pipeline to evaluate hyperbolic latent navigation for multi-step reasoning; target submission to **NeurIPS 2026**.

Nanjing Medical University, Jiangsu, China

Feb 2025 – Sep 2025

Research Assistant | AI for healthcare

Advisor: Mingli He

Project: Machine Learning and Global Epidemiological Modeling for Stroke Risk and Outcome Prediction

- Served as **co-corresponding author**, providing methodological oversight and analytical guidance to an interdisciplinary research team.
- Contributed to the **design and evaluation** of an interpretable XGBoost model for disease progression prediction and clinical decision support (ROC-AUC > 0.90).
- Guided **statistical and machine learning modeling**, applying generalized additive models (GAMs) to study nonlinear socioeconomic-risk interactions in GBD-based stroke research.
- Contributed to **manuscript preparation** and provided continued technical input during peer review and revision.

Fudan University, Shanghai, China

Nov 2024 – Apr 2025

Research Assistant | Data Science

Advisor: Chaoyan Yue

Project: Global Epidemiological Modeling of Ischemic Stroke in Young Adults (1990–2021)

- Analyzed global ischemic stroke trends (1990–2021) in young adults using GBD 2021 data, employing **Joinpoint regression** and **Bayesian APC models** to forecast disease burden.
- Contributed to **data cleaning and statistical analysis**, and assisted with figure preparation and drafting parts of the methods and results sections to ensure clear and consistent data presentation.

PUBLICATIONS

* Corresponding author.

Conference Papers

- [C.1] **He, Y.**, Wen, Y., Wang, X., & Ma, T.* (2026). MTS-JEPA: Multi-Resolution Joint-Embedding Predictive Architecture for Time-Series Anomaly Prediction. *arXiv:2602.04643*. Under review at *ICML 2026*.

Journal Articles

- [J.1] Li, J., Chang, H., Du, S., et al., **He, Y.***, & He, M.* (2025). Development and validation of a web-based machine learning model for predicting early neurological deterioration following stroke thrombolysis: A multicenter study. *Journal of Medical Internet Research*, 27, e77858. 10.2196/77858.
- [J.2] Li, J., Zhang, C., Zhang, H., Gu, X., **He, Y.***, & He, M.* (2025). Socioeconomic gradients in inflammatory risks and global stroke mortality. Under review at *The Lancet Neurology*.
- [J.3] He, M., Zhang, S., Liu, X., **He, Y.**, Gu, X.* , & Yue, C.* (2025). Global, regional, and national epidemiology of ischemic stroke in young adults, 1990–2021. *Journal of Neurology*, 272(5), 354. 10.1007/s00415-025-13082-4.

SKILLS

Programming Languages: Python, Java, C/C++, Assembly, R, SQL, MATLAB, JavaScript, HTML/CSS

Expertise: Deep Learning, Machine Learning, Statistical Modeling, PyTorch, TensorFlow, scikit-learn, Transformers

DevOps & Cloud: Git, Kaggle, Linux, Docker, AWS (EC2, S3, Lambda), Google Cloud, Vast.ai

Web Development: Flask, FastAPI, React.js, Node.js, RESTful API design, PostgreSQL, JSON

Languages: Chinese (Native), English (Fluent)